IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE PART III/IV: MEng, BEng and ACGI

ELECTRICAL ENERGY SYSTEMS

Wednesday, 11 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer any 2 out of 3 questions from Section A and answer 2 out of 3 questions from Section B.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, F. Teng

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Section A: Answer any two questions out of Q1 to Q3

1. **Question on power system design and frequency regulation**

a) Assume that there are n identical generators, each with capacity of 50 kW and availability of 0.98.

i) The system design standard requires that the probability of meeting P kW peak demand is equal or greater than 0.99. As a system design engineer, formulate a general optimization problem to determine the minimum number of the generators n_{min} that meet this standard (P is a parameter in the optimization). Base on the formulated optimization problem, Find n_{min} when $P = 100$.

[6 marks]

ii) Assume that the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW and the reliability criterion requires that the probability of meeting peak demand is equal or greater than 94%. At this moment the number of generators $n = 2$, Calculate how many households can be supplied? Now, the area has a plan for new property development and the total household number is increased **to** 58. Calculate and determine if the current system still meets the reliability criterion. If not, how many more generators are needed?

[4 marks]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore a new service called Dynamic Containment (DC) is designed in the UK. DC can effectively reduce the need for the regular frequency response (R) to maintain the frequency nadir as required, following the rule below:

$$\left(\frac{H}{f_0} - \mathrm{DC} \times \frac{T_{\mathrm{DC}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_L - \mathrm{DC})^2 \times T_d}{4 \times \Delta f_{\max}} \tag{1.1}$$

Nomenclature:

H: System level of inertia (MW-s)
R: Amount of frequency response scheduled (MW)
DC: Amount of DCscheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
f_0: Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$: Maximum admissible frequency deviation (Hz)
$\mathrm{T_d}$: Delivery time of frequency response (s)
T_{DC}: Delivery time of DC (s)

Assumptions:

- The amount of frequency response: $R = 2000$ MWs,
- The cost of frequency response: \$5/MW,
- The largest generation unit in the UK: 1.8 GW,
- Other parameters: $f_0 = 50$ Hz, $\Delta f_{\max} = 0.8$ Hz, $T_d = 10$ s, $T_{\mathrm{DC}} = 0.5$ s.

i) Based on (1.1), comment on how H, R, DC, T_d, T_{DC}, P_L affect the frequency nadir (the largest frequency reduction).

[2 marks]

ii) Calculate the minimum required inertia in the system to maintain the frequency nadir constraint when DC =200 MW and 300 MW respectively.

[4 marks]

iii) Assume that the frequency response is optimally procured at the current price. Determine the economic value of H in terms of $\$/\text{GWs}$ on maintaining the frequency nadir constraint, in the case of DC = 300 MW.

[4 marks]

2. **Question on per unit system**

a) How to choose the base voltage for the primary and secondary sides of an ideal transformer? What benefit does it bring for the per unit system analysis and why?

[4 marks]

b) Consider the high-voltage transmission circuit in Figure 2.1. The current at the receiving end lags V_r by 45° and $V_r = 1\angle 0°$ p.u., $|V_s| = 1.3$ p.u..

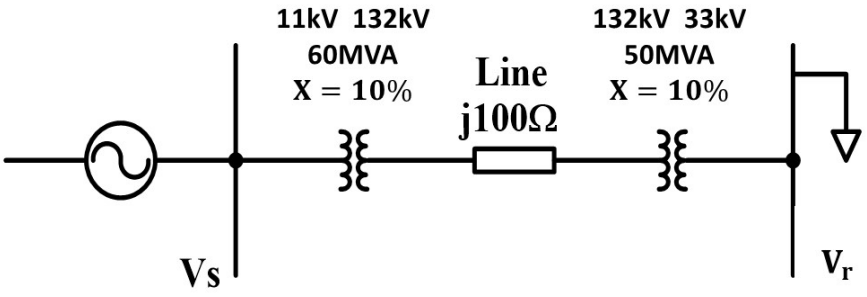


Figure 2.1: Transmission circuit with powers specified at both sending and receiving ends

i) Calculate the per unit values of the quantities (three impedances) in the figure, with properly selected bases.

[4 marks]

ii) Draw the phasor diagram and calculate V_s and the magnitude of the line current in per unit.

[12 marks]

3. **Question on power flow analysis and Gauss-Seidel method**

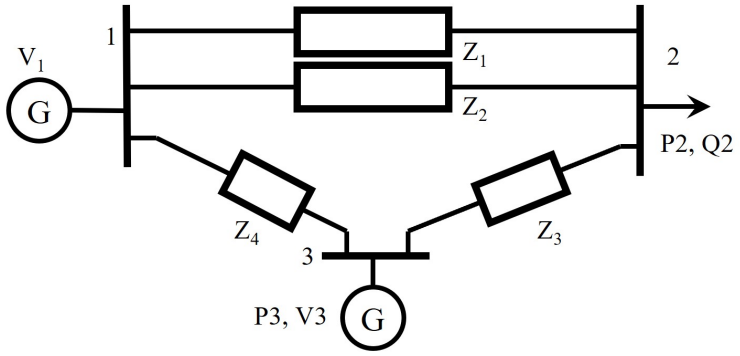


Figure 3.1: A three-bus power network

A three-bus power network is presented in Figure 3.1.

a) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3, Z_4.

[5 marks]

b) When applying the Gauss-Seidel method to solve the load flow problem in a general multi-bus system, the bus voltages are updated in each iteration.

i) Derive the bus voltage update law for V_2 and V_3 in Figure 3.1 based on the node quantities given in the figure and the elements in the admittance matrix (assume V_1 is known). Explain why the bus voltages cannot be solved directly.

[3 marks]

ii) Based on the derived voltage update law, properly define the vector $\mathbf{x}$ and the multidimensional function $\mathbf{h}$, such that the Gauss-Seidel method can be generalized from the load flow problem to a nonlinear equation $\mathbf{x} = \mathbf{h}(\mathbf{x})$. Explain how to perform the Gauss-Seidel method in this general case.

[5 marks]

c) DC power flow approximation

i) The power flow between two buses can be written as: $P_{km} = \frac{V_k V_m \sin\theta_{km}}{X_{km}}$, $Q_{km} = \frac{V_k^2 - V_k V_m \cos\theta_{km}}{X_{km}}$, after neglecting line losses and shunt admittance. Comment on how the bus angle difference and voltage magnitude influence the active and reactive power respectively, by evaluating the sensitivities of P_{km} and Q_{km} with respective to θ_{km} and V_k at $\theta_{km} = 0$ (four quantities in total).

[2 marks]

ii) To simplify the power flow analysis, the so-called DC power flow approximation can be used, where the relationship between the active power and the bus angle difference is described in a linear manner. In this case, we can assume that $V_k \approx V_m \approx 1$p.u. and $\sin\theta_{km} \approx \theta_{km}$. Rewrite the expression of P_{km} as a function of θ_{km}.

Similar to $\mathbf{I} = Y\mathbf{V}$ as derived in a), which describe the relationship between node current injection and the bus voltage through the admittance matrix, the relationship between the node active power injection and the bus voltage angle can also be describe in a linear fashion

through the B matrix in the DC power flow approximation. Derive the expression the matrix B as in $\mathbf{P} = B\boldsymbol{\theta}$, where $\mathbf{P}$ is the vector of bus active power injection P_k and $\boldsymbol{\theta}$ is the vector of bus voltage angle θ_k. Formulate matrix B as a function of line reactance X_{km}.

[5 marks]

Section B: Answer any two questions out of Q4 to Q6

4. **Question on balanced fault analysis**

a) Explain the following statements:

i) The fault response of an inverter-based resource is different from that of a synchronous generator.

[2 marks]

ii) It is reasonable to neglect the line currents before a fault to simplify fault calculations.

[2 marks]

iii) In a closed electrical circuit that contains both finite resistance and electromotive force (EMF), the magnetic flux linkage cannot undergo an instantaneous change.

[2 marks]

iv) High shares of inverter-based resources such as wind, and solar photovoltaic in Great Britain could result in a decrease or an increase in fault level.

[2 marks]

b) A power grid with only synchronous generation has N_b number of buses and N_l number of transmission lines. A balanced fault occurs at bus k with a fault impedance Z_f. Derive expressions for the following, utilizing the bus impedance matrix Z_b, pre-fault voltage V_k^0 at bus k, and Z_f. The element Z_{mn} in row m and column n of Z_b relates to bus m and n of the power grid.

i) Fault current I_f at bus k

[4 marks]

ii) Voltage at a non-fault bus b, V_b^f during the fault

[2 marks]

c) Consider the bus impedance matrix in (4.1) for a system with 4 buses where each row and column represents the corresponding bus number.

$$Z_b = \begin{bmatrix} j0.20 & j0.10 & j0.05 & j0.03 \\ j0.10 & j0.25 & j0.04 & j0.02 \\ j0.05 & j0.04 & j0.22 & j0.01 \\ j0.03 & j0.02 & j0.01 & j0.18 \end{bmatrix} \text{ p.u.} \qquad (4.1)$$

Assume a three-phase fault at bus 3 with a fault impedance $Z_f = 0$. Neglect pre-fault currents and assume that the pre-fault voltages at all buses are 1.0 p.u.

i) Calculate the fault current in per unit (p.u.).

[2 marks]

ii) Find the voltage (in p.u.) at bus 1, 2, and 4 during the fault.

[2 marks]

iii) Determine the Thevenin equivalent impedance at bus 2 and 4.

[2 marks]

5. **Question on unbalanced fault analysis**

a) Based on the fault current and voltage in the phase domain, describe how the three sequence networks should be interconnected to analyse a single line-to-ground (LG) fault on phase A with a fault impedance of Z_f to ground.

[5 marks]

b) A balanced three-phase power supply delivers energy to a delta-connected three-phase load which draws a balanced line current of 580 A. A sudden open circuit occurs in phase A causing the current in phase A to zero. Using the current in Phase C as the reference with 0° phase angle, determine the magnitude and phase angle (in degrees) for the positive-sequence (A-B-C), negative-sequence (A-C-B), and zero-sequence components of the resulting unbalanced current. For your answers limit the phase angles within $\pm 180^\circ$.

[5 marks]

c) A 66 kV three-phase line is connected to an 11 kV generator through an 11/66 kV transformer. The 11 kV side of the transformer is delta-connected while the 66 kV side is star-connected with solidly grounded neutral. The positive, negative, and zero sequence reactance of the generator, transformer, and line are given in Table 5.1.

Component	Positive Seq.	Negative Seq.	Zero Seq.
Generator	$X_1'' = 0.01$ p.u.	$X_2'' = 0.01$ p.u.	$X_0'' = 0.005$ p.u.
Transformer	$X_1 = 0.05$ p.u.	$X_2 = 0.05$ p.u.	$X_0 = 0.05$ p.u.
Line	$X_1 = 0.1$ p.u.	$X_2 = 0.1$ p.u.	$X_0 = 0.22$ p.u.

Table 5.1: Sequence reactance for generators, transformers, and line.

Considering an unbalanced fault at the other end of the transmission line, determine the following, neglecting the fault impedance and any resistance. Use a system base power of 100 MVA for your calculations.

i) The fault current (in kA) for a line-to-ground (LG) fault on phase A.

[5 marks]

ii) The voltage (in kV) in phase B at the fault point for the same line-to-ground (LG) fault on phase A.

[5 marks]

6. **Question on stability analysis**

a) Define transient stability in power systems.

[2 marks]

b) State the two main factors that influence transient stability in power systems and give an example of each.

[3 marks]

c) Explain, using a diagram, the factors that limit the power transfer capacity of uncompensated AC overhead transmission lines as a function of line length.

[5 marks]

d) Mention two important considerations that limit the use of high series compensation levels in long transmission lines.

[2 marks]

e) A 400 MVA, 50 Hz round-rotor synchronous generator has a synchronous reactance of 0.15 p.u. and an inertia constant of 5 MW-s/MVA. The generator is connected to a large power grid through a transformer and overhead line with a combined reactance of 0.45 p.u. on a base of 400 MVA. The internal voltage of the generator and the voltage at the point where the overhead line connects to the large power grid are both tightly regulated at 1.0 p.u. The generator supplies 500 MW to the grid. Neglect resistance and consider a 400 MVA system base. For a three-phase temporary fault on the transmission line calculate the critical clearing time (in seconds) for the following cases assuming zero power output from the generator during the fault:

i) without any line compensation.

[5 marks]

ii) with 40% series compensation on the line.

[3 marks]